# Design Problem

A1

**The Problem**

Joanna Chan (*Fig 1*) is a mother who constantly needs to shop for her family. She shops 3-4 times a week from the Sheng Siong approximately 1 km away. Joanna often feels **strains** on her hand and arm when carrying bags from the supermarket to her car and from the car back up to her apartment. These strains may vary depending on the **weight** of the shopping bags and the **distance** varies depending on where she parks.

In order to come up with a suitable solution, research will have to be done to gather more information about the client's shopping habits and anthropometrics

*Fig 1: Photo of client*

**Primary research**

*Fig 2: Client's grip*


*Fig 3: Client's primary bags*

| | |
|---|---|
| **Bag weights** | 58.4, 58.5, 79.0 |
| **Grocery weight range** | Up to 5kg |
| **Number of bags** | 1-3 bags |
| **Approximate distance travelled by foot** | 30 meters |

*Fig 4: Information on shopping habits*

Although the travel distance by foot is not long, as the weight of the groceries increase, the **strain** on the fingers will also increase, as seen in *Fig 2,* the client's grip.

**Anthropometrics of client**

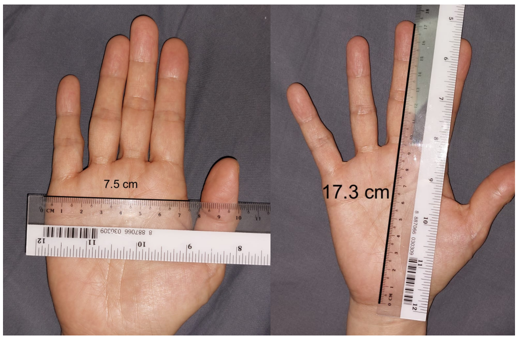



*Fig 5: Size of hand*

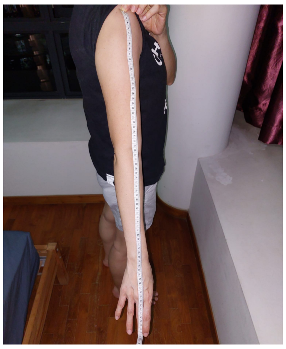


*Fig 6: Length of arm*

| | |
|---|---|
| **Hand breadth (cm)** | 7.5 |
| **Hand length (cm)** | 17.3 |
| **Arm length (cm)** | 72.5 |
| **Height (cm)** | 159 |

*Fig 7: Height of client*

Information from *Fig 5, Fig 6* and *Fig 7* will let me know **limitations** for the **size** of the product and what size will provide the client the most comfort

**Secondary research**

Height (female)

| | |
|---|---|
| **5th percentile** | 150cm |
| **50th percentile** | 160cm |
| **95th percentile** | 173cm |

Client: 46th percentile

Hand length (female)

| | |
|---|---|
| **5th percentile** | 16.5cm |
| **50th percentile** | 18.5cm |
| **95th percentile** | 19.7cm |

Client: 23rd percentile

Hand breadth (female)

| | |
|---|---|
| **5th percentile** | 7.3cm |
| **50th percentile** | 7.9cm |
| **95th percentile** | 8.6cm |

Client: 20th percentile

Arm length (female)

Arm length is approximately proportional to height

Client: 46th percentile

# Design Brief

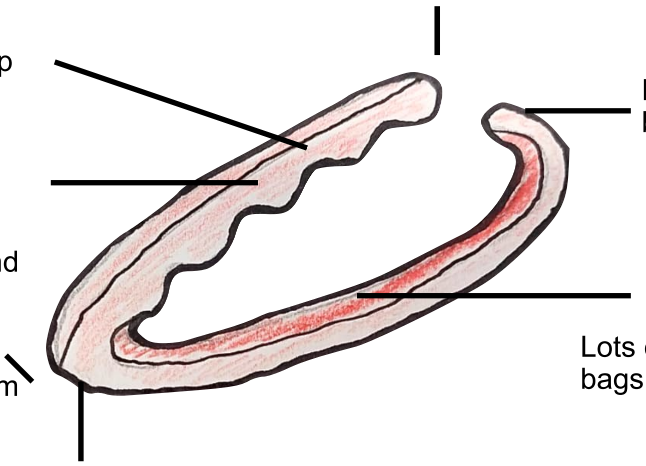
A2

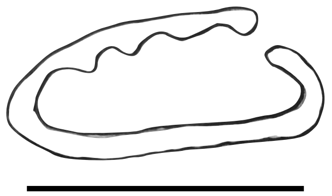
I propose to create a product that efficiently carries up to 4 handbags with comfort for my client. The product will **reduce stress** on the client's arms and be able to hold shopping bags of at least **7kg**. For the product to serve its purpose, it must be **light**, **portable** and **small**. In order for the product to suit my client, it must be **no taller than 20cm** or else the bags may scrape the floor. The product must be **easy to use** and be **stored easily** when not in use.

# Conceptual Designs

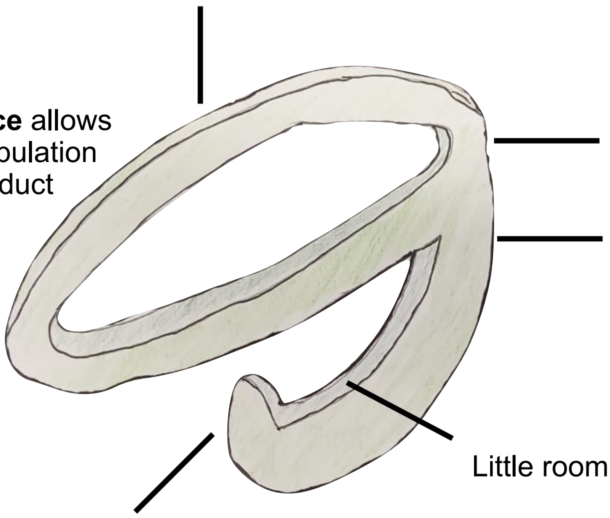
B1

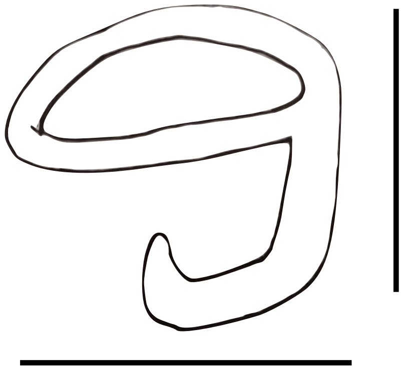
**Design 1**



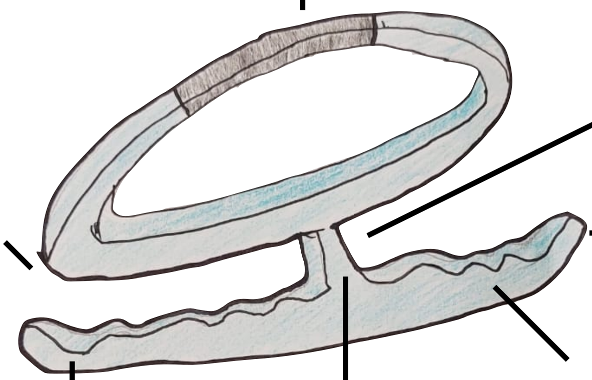



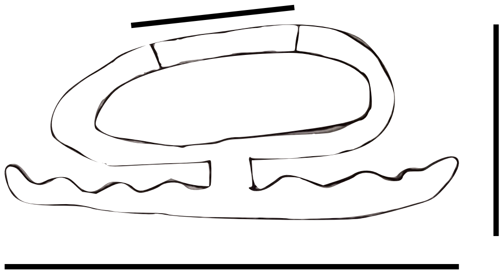

**Design 2**



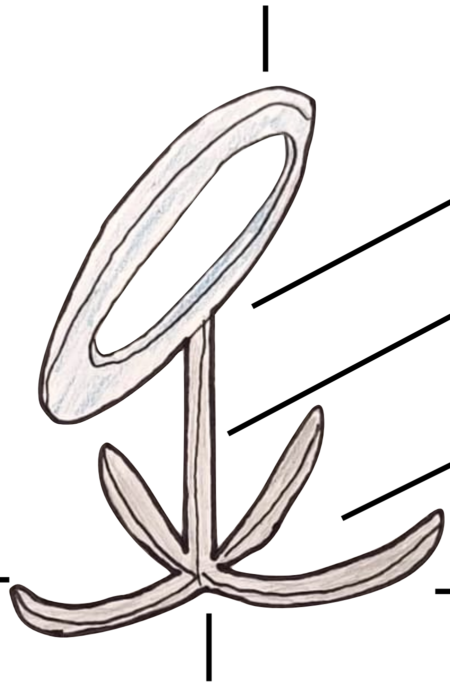



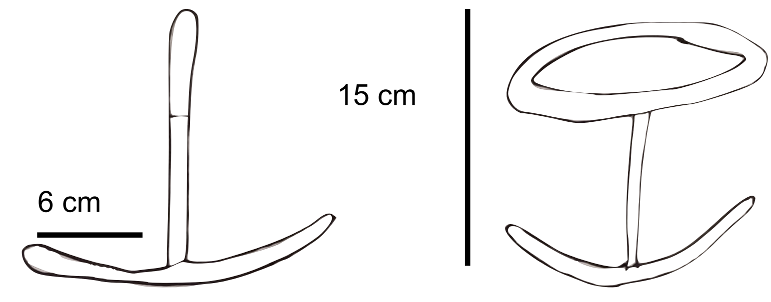
**Design 3**




**Design 4**

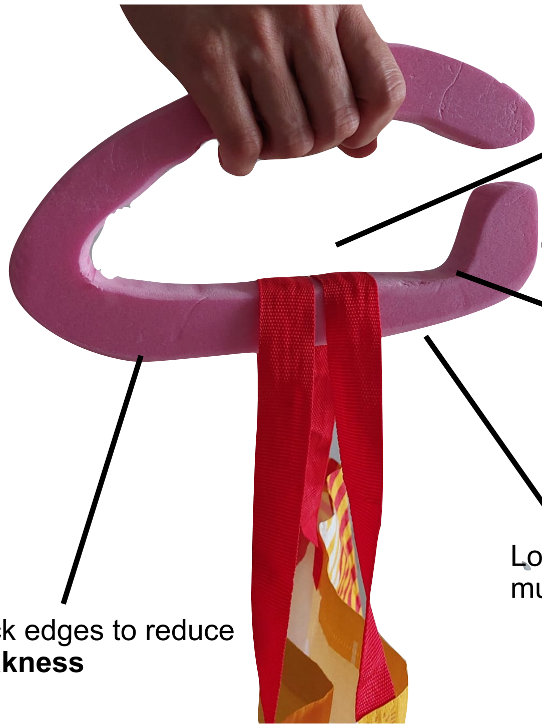


# Development of conceptual designs

B2

## Development of idea 1



High **clearance** for user

Carrier may be unnecessarily heavy

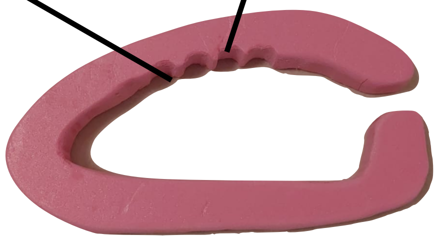
Could be a bit big for user, causing **discomfort**

Lots of room for multiple bags

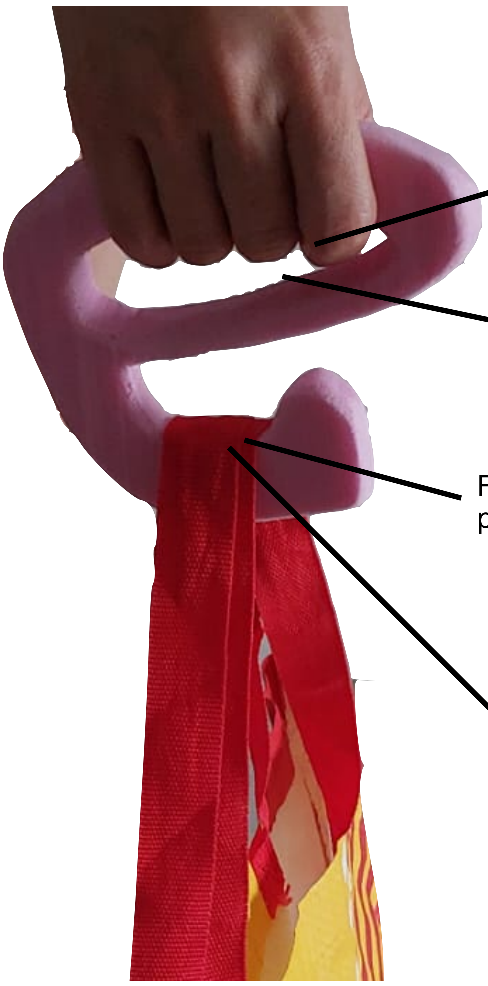
Thick edges to reduce **weakness**

Contours match user's finger size but not other population percentages

Contour for more effective grip and clear indication of handle



## Development of design 2

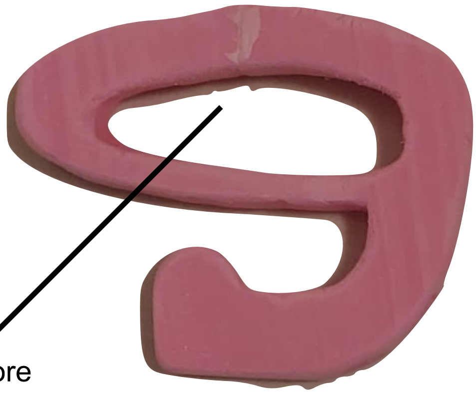


Perfect size for hand (**20th percentile**)

Reduced space. Extra room is unnecessary and causes **weakness**

Flattened out surface to provide more space for bags

Bags are all bunched in one place (**higher stress**)



Reduced grip size for more control of carrier

Increased thickness to **prevent snapping**

B2

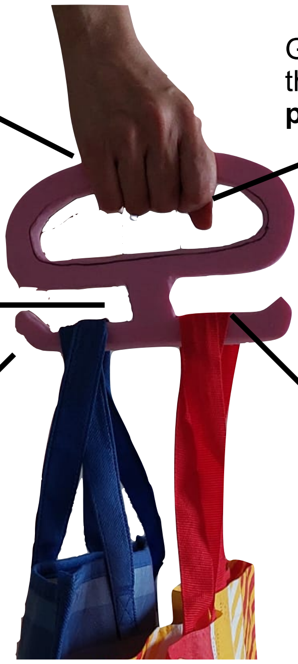
**Development of idea 3**

Removed foam handle

Foam handle will either cause **weakness** because of indent or grip will be too large for client

Increased thickness of neck

Increased degree of hooked ends to compromise for smoothened surface



Grip size still accounts for the **majority of the population**

Shrunk grip size as previous grip size was too wide

Smoothened surface to avoid **weaknesses**

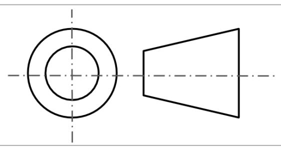
## Justification of final design

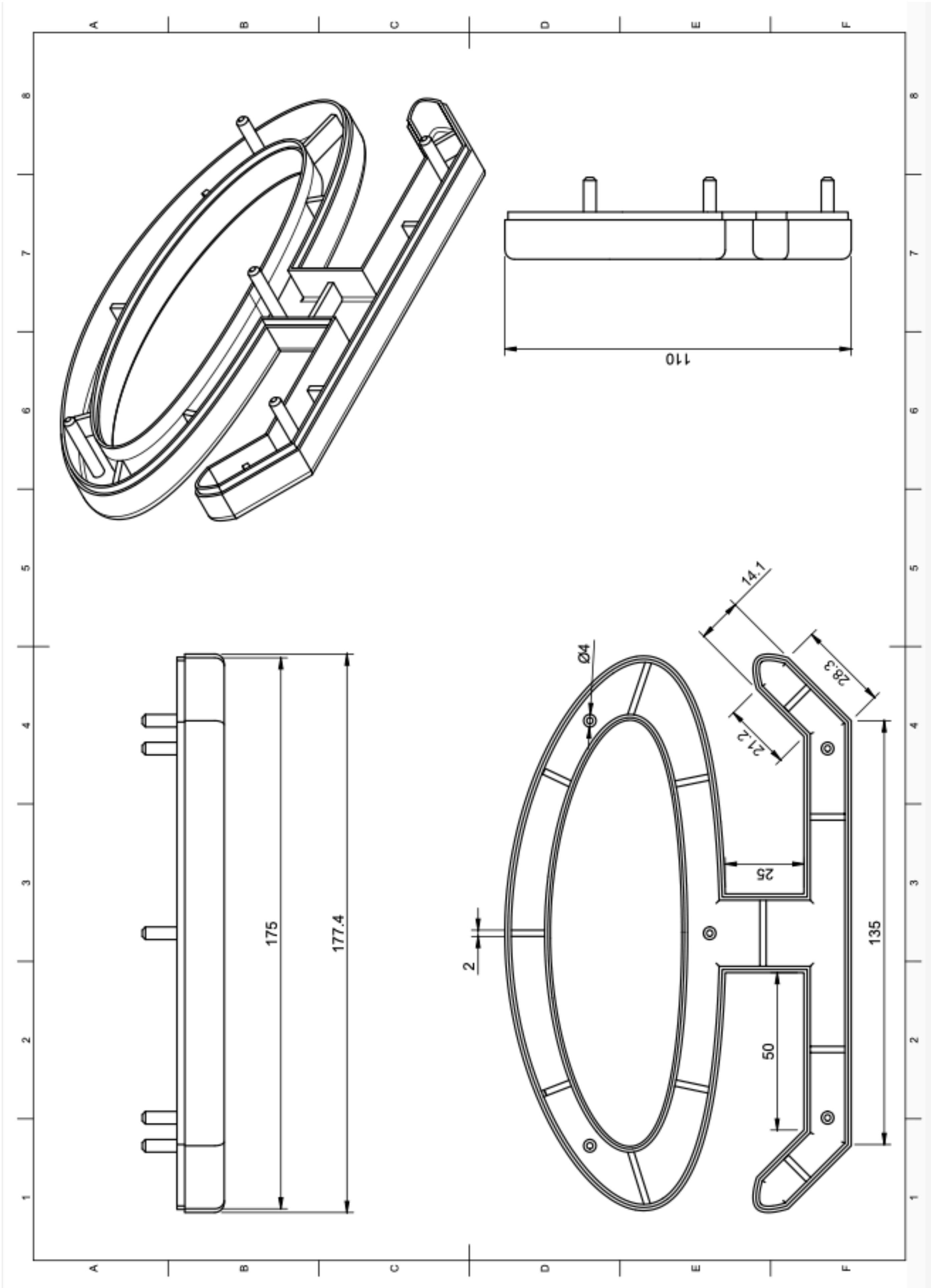
B3

I have chosen to proceed with design 2 as I feel that this design most effectively matches my client's needs. The bag carrier allows **multiple bags** to be carried at once and has 2 sides to maintain a **balance** and to not put the design under too much **stress**. There is lots of room for the grip which will provide **comfort** for the client. In addition, the wide grip will be a blue to account for a population range from the **5th to 95th percentile**. The design has no sharp edges that can harm the client and **smooth edges** will provide **comfort** for the client's hand. Furthermore, the design will be made with ABS which has a **high strength to weight ratio**. The simplified design will make it so that it has a **sleek** look, which will make it **easy to understand**.

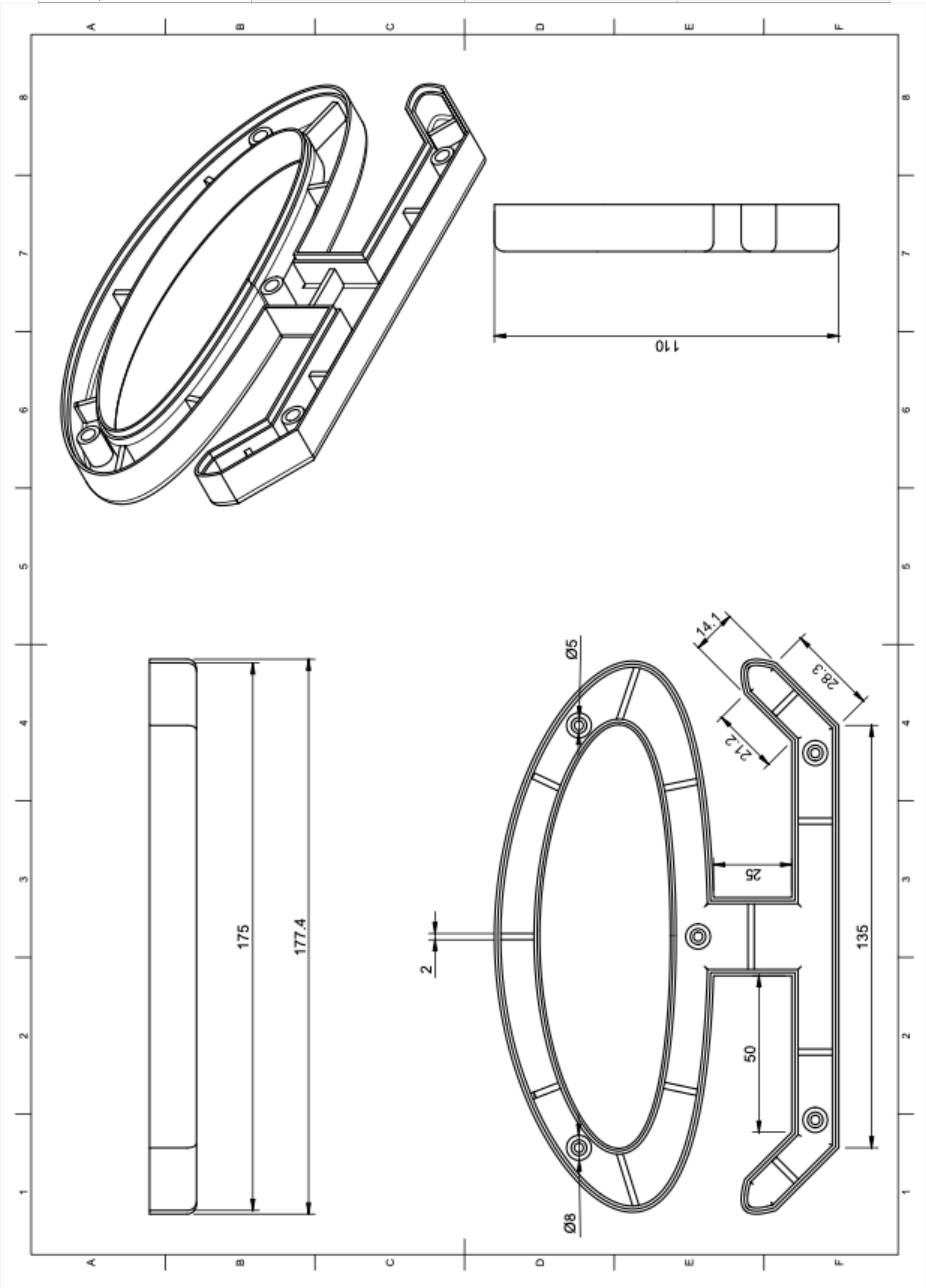
# Design Proposal

C2

| | | | |
|---|---|---|---|
|  | Component name: Male part | Materials: ABS | Scale: 1:1 |
| | Part number: 1 | Quantity: 1 | Unit: mm |

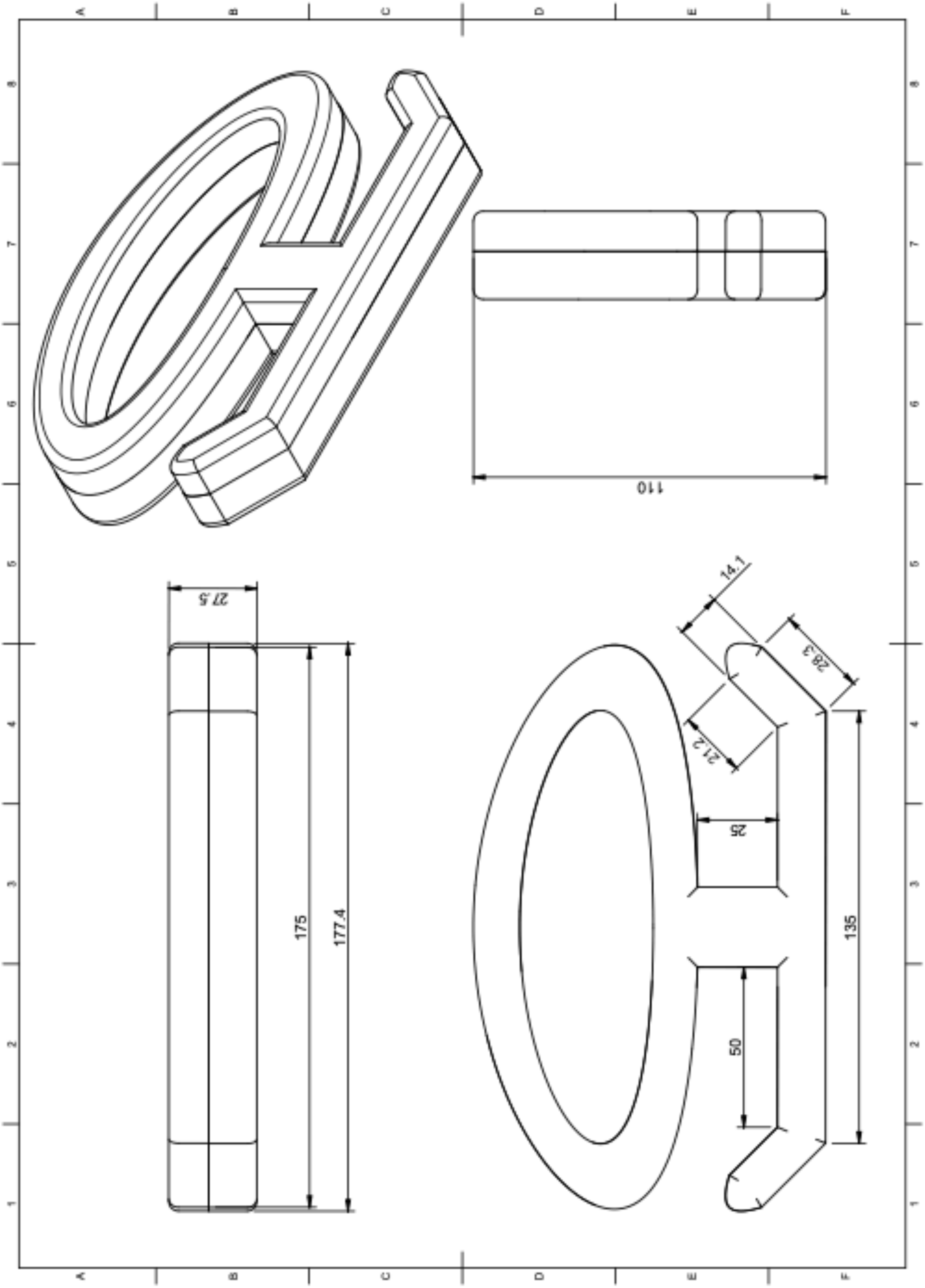


| | | | |
|---|---|---|---|
| | Component name: Female part | Materials: ABS | Scale: 1:1 |
| | Part number: 2 | Quantity: 1 | Unit: mm |



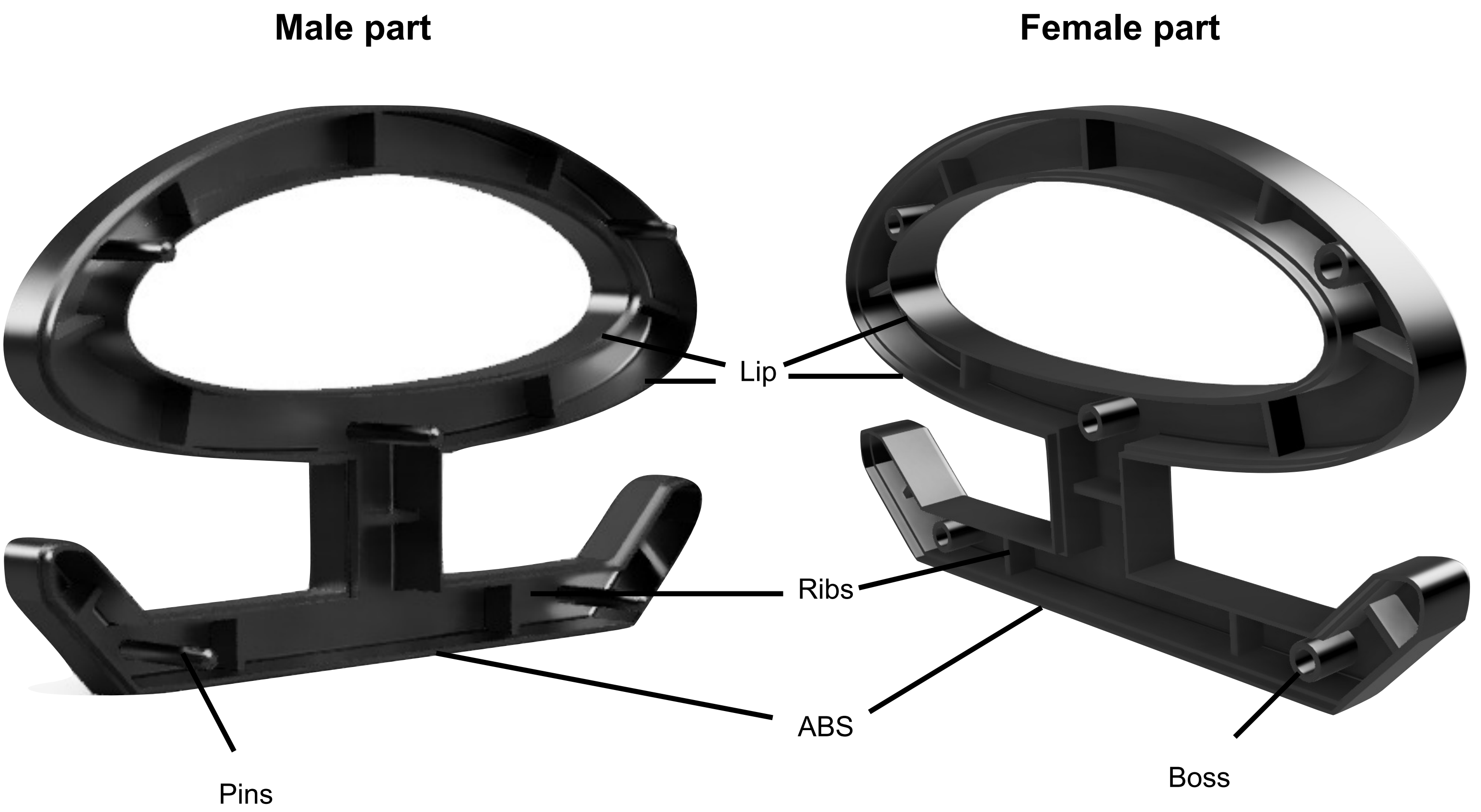
C2

| | | | |
|---|---|---|---|
| | Component name: Combined product | Materials: ABS | Scale: 1:1 |
| | Part number: 1+2 | Quantity: 1 | Unit: mm |



# Commercial Production






| Part | Purpose |
|---|---|
| Boss | Secures components in place with pins |
| Pins | Secures components in place with boss; fits into the boss |
| Ribs | Stabilises the components and prevents bending |
| Lips | An offset on both sides that creates a tight but secure fit with both components |

**Purpose and properties of ABS**
ABS is a well suited material for commercial production. Below are properties that make ABS well suited for this:

- Melting point is 210°C-260°C
- Molding temperature is 45°C-80°C
- Drying temperature is 80°C
- Drying time is 2-9 hours
- Rigid, lightweight, durable

# Stress analysis



**Load and constraints**
For this FEA analysis, 7 kgs of load was placed as shown in *Fig 8*. *Fig 9* represents the constraints (the part of the product that won't move when load is applied) at the handle as this is where the carrier is held. Note that the constraints are not perfect and should only be where the hand is gripped, not the whole inner oval. Due to these new constraints, the displacement and stress shown in the analysis will only be at the bottom half of the carrier.

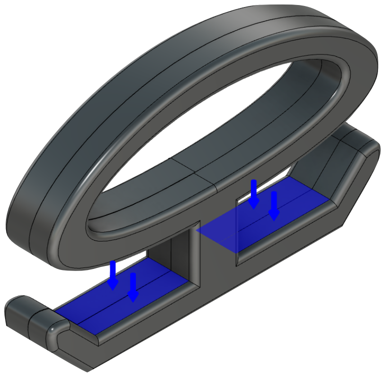


*Fig 8: Load*

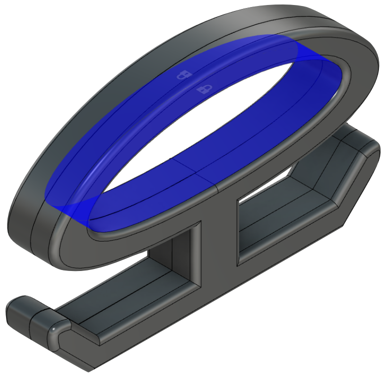


*Fig 9: Constraints*

**Displacement**
The displacement test, as shown in *Fig 10*, represents how much the product is displaced when load is applied. Looking at the results, it can be seen that the most displacement is at either hooked ends which makes sense since the ends have less support and are weaker than the center. To strengthen the structure, a possible idea would be to shorten the base on either side or thicken the central support

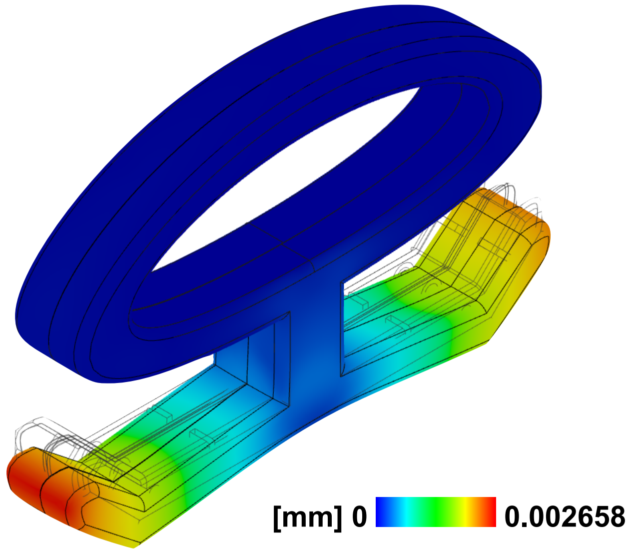



*Fig 10: Displacement*

**Stress**
The von mises stress test, as shown in *Fig 11*, which parts of the structure is under the most stress. This test considers whether a part of the structure will yield or fracture. As shown by *Fig 11* it can be seen that the area around the bottom center is under the most stress. To reduce stress in this area, a possible solution would be to thicken the base and the center.

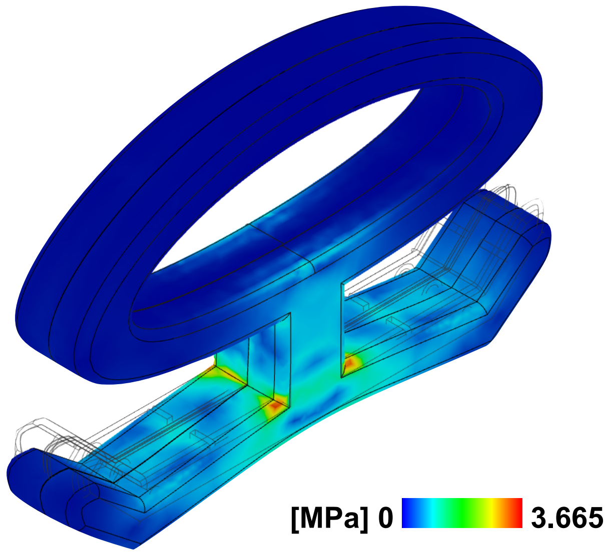



*Fig 11: Stress*

# Final product






# Sources

https://www.hermanmiller.com/research/categories/white-papers/the-evolution-of-anthropometrics-and-user-control/#:~:text=To%20further%20illustrate%3A%20A%2095th,229%20pounds%20(103.87%20kg).

https://www.gripboard.com/index.php?/topic/40936-interesting-hand-size-chart/

https://www.fictiv.com/articles/abs-injection-molding-material-properties-and-processing-considerations